

Phase 1 Diagnostics Report

Hidden Markov Model (Market Painter) - Goodness-of-Fit, Assumption Checks, and Implications

Executive summary

This report summarizes statistical diagnostics for Phase 1 (Market Painter) based on the produced outputs: PIT calibration tests, serial dependence tests, emission distribution checks, numerical stability checks, and parametric-bootstrap likelihood-ratio tests for the number of regimes. The results provide strong evidence that regime structure is warranted, but also identify material misspecification under Gaussian emissions and residual time-series dependence. These findings motivate targeted improvements; depending on intended use (state labeling vs full probabilistic modeling), improvements may range from heavier-tailed emissions and switching volatility to an HSMM with covariate-driven duration exit.

Training PIT summary

Model	Training PIT count	Mean PIT u	Std PIT u	Mean z	Std z
HMM q=2	2317	0.450	0.289	-0.174	1.013
HMM q=4	2317	0.431	0.293	-0.249	1.040

Diagnostics methodology

The diagnostics focus on (i) calibration of the one-step-ahead predictive distribution, (ii) residual dependence, (iii) emission distribution adequacy under Gaussian assumptions, (iv) numerical stability of estimated covariance matrices, and (v) evidence for the number of regimes via parametric bootstrap likelihood-ratio tests.

Probability integral transform (PIT) calibration (Kolmogorov-Smirnov)

For each training observation, compute a PIT value $u_t = F(y_t | F_{t-1})$, where F is the model's one-step-ahead predictive CDF. Under correct specification, PITs are i.i.d. $\text{Uniform}(0,1)$. We test $H_0: u_t \sim \text{Uniform}(0,1)$ using the Kolmogorov-Smirnov test.

Independence of residuals (Ljung-Box on z and z^2)

Transform PITs into $z_t = \Phi^{-1}(u_t)$. Under correct specification, z_t should be approximately i.i.d. $N(0,1)$. We use Ljung-Box tests on z_t and on z_t^2 at lags 10 and 20. H_0 : no autocorrelation up to the specified lag.

Conditional heteroskedasticity (ARCH LM)

Using z_t , apply the ARCH Lagrange Multiplier test. H_0 : no ARCH effects up to the chosen lag. Rejection suggests remaining volatility clustering not captured by the model.

Emission distribution adequacy (Jarque-Bera within regimes)

Phase 1 uses Gaussian emissions. Within each inferred regime, Jarque-Bera tests check normality of each series. H_0 : regime-conditional returns are normally distributed (zero skewness and normal kurtosis).

Numerical stability (covariance condition number)

For each state, compute the condition number of the estimated covariance matrix. Very large values (or infinity) indicate near-singular covariance, which undermines density evaluation and may signal weak state identification.

Number of regimes (parametric bootstrap LR tests)

Likelihood-ratio (LR) tests in HMMs are non-standard due to non-regularity (label switching, boundary issues). We therefore use a parametric bootstrap: fit null and alternative models, compute LR_{obs} , simulate data under the null, refit both models on each bootstrap sample, and compute a bootstrap p-value as the fraction of $LR_b \geq LR_{\text{obs}}$.

Results

Table 1. Calibration and dependence tests (training)

Model	Train LL	KS p-value (PIT)	LB p (z) lag10	LB p (z) lag20	LB p (z^2) lag10	LB p (z^2) lag20	ARCH LM p-value
HMM_q2	15248.61	9.51e-14	8.26e-44	7.27e-79	0.0394	0.0023	0.0582
HMM_q4	54382.57	3.01e-25	4.11e-40	3.12e-74	0.0079	1.36e-06	0.0176

Key findings from Table 1

- PIT uniformity is rejected for both $q=2$ and $q=4$ (very small KS p-values), indicating predictive miscalibration under Gaussian emissions.
- Ljung-Box p-values for z at lags 10 and 20 are effectively zero for both models, indicating material remaining serial dependence.
- Ljung-Box on z^2 and ARCH LM tests indicate remaining volatility clustering, more pronounced for $q=4$ (ARCH LM p-value 0.0176).

Figure 1. PIT histogram (training) - HMM q=2

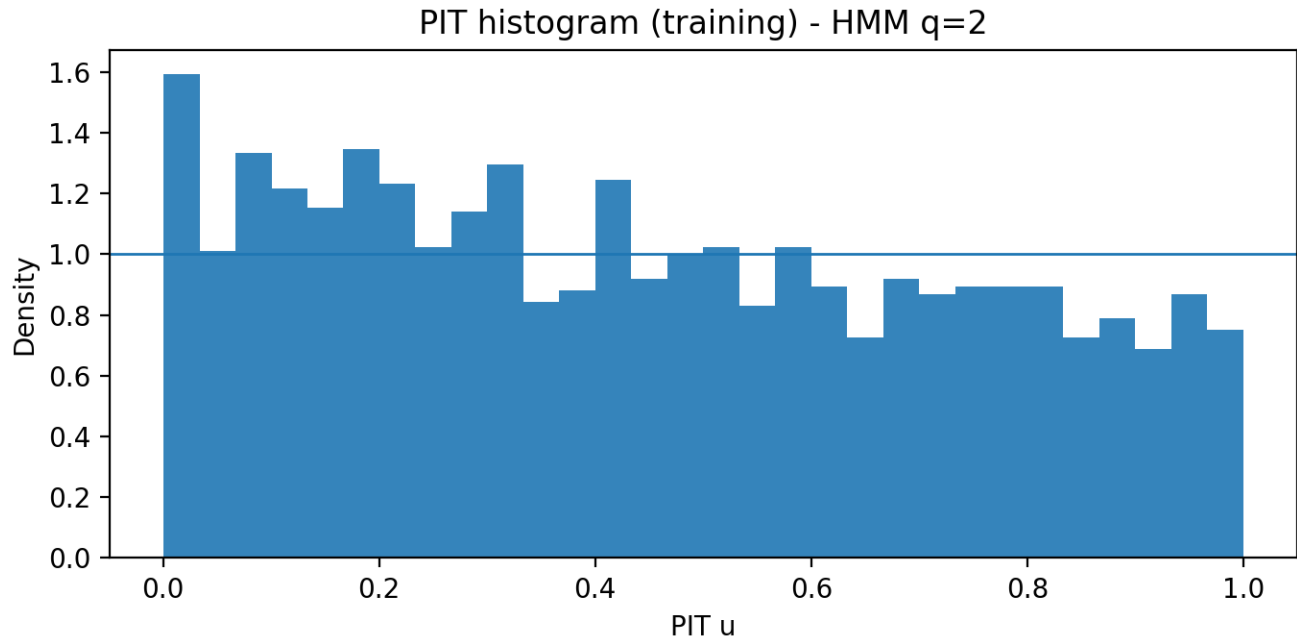
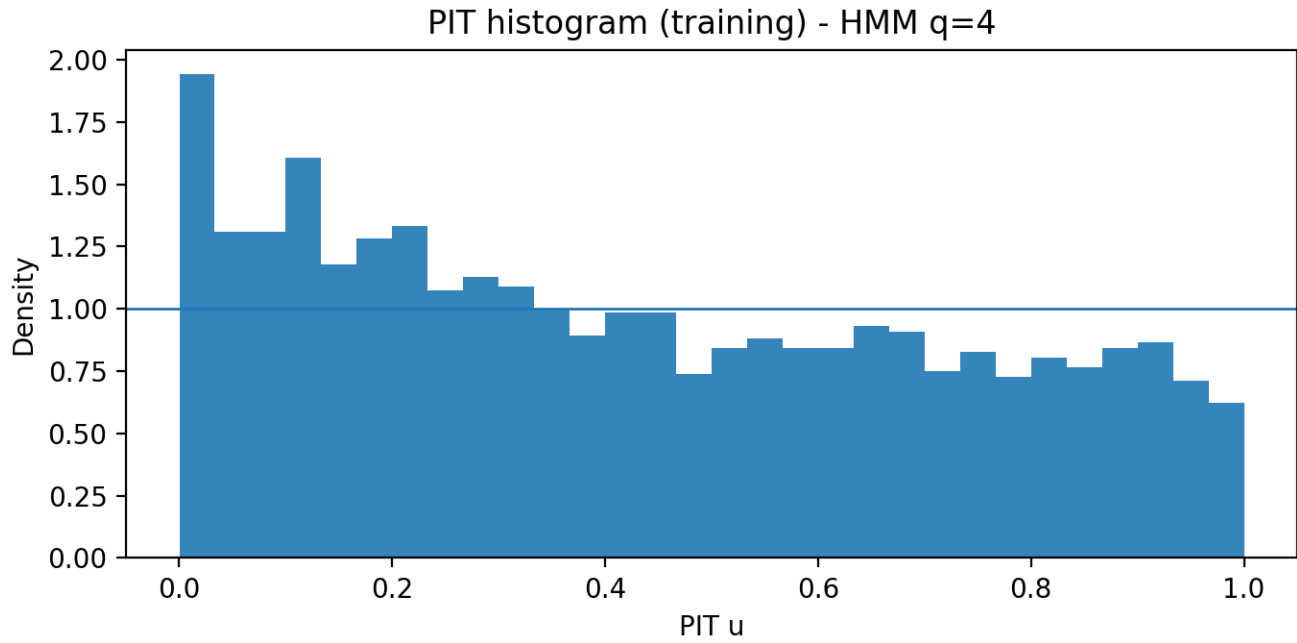


Figure 2. PIT histogram (training) - HMM q=4



Residual diagnostics via transformed PITs

Q-Q plots assess distributional fit of $z = \Phi^{-1}(u)$ against $N(0,1)$.

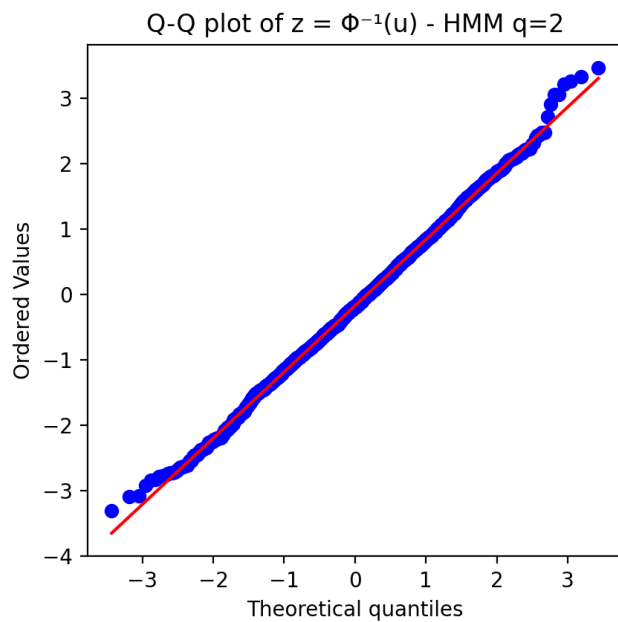


Figure 3. Q-Q plot of z - HMM $q=2$

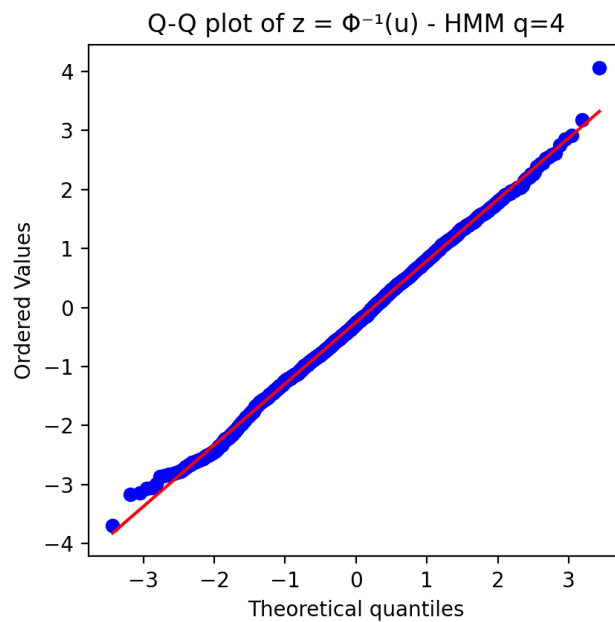
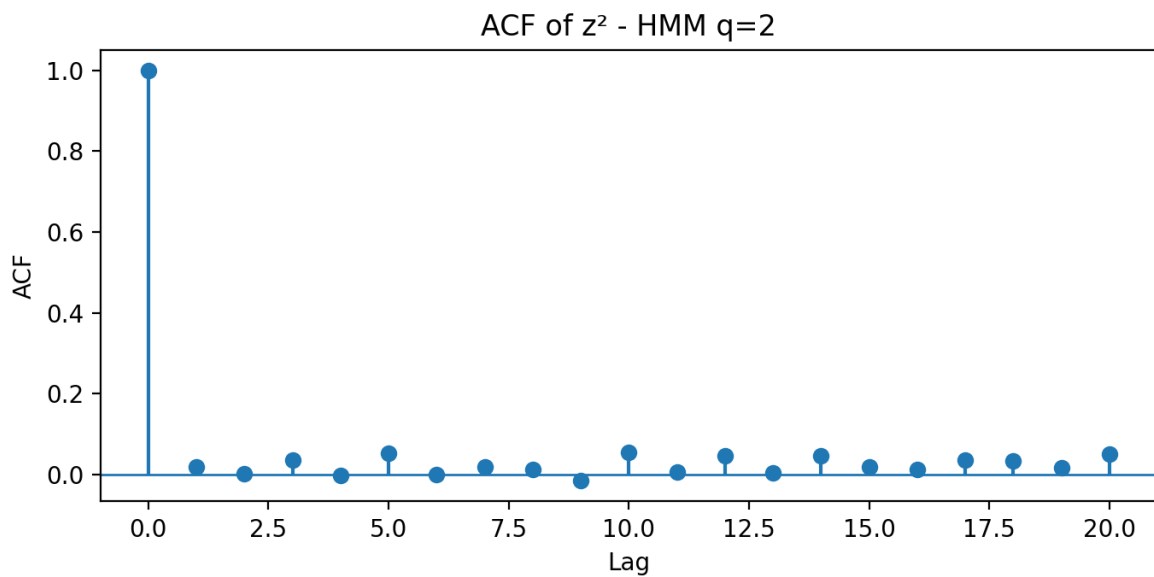
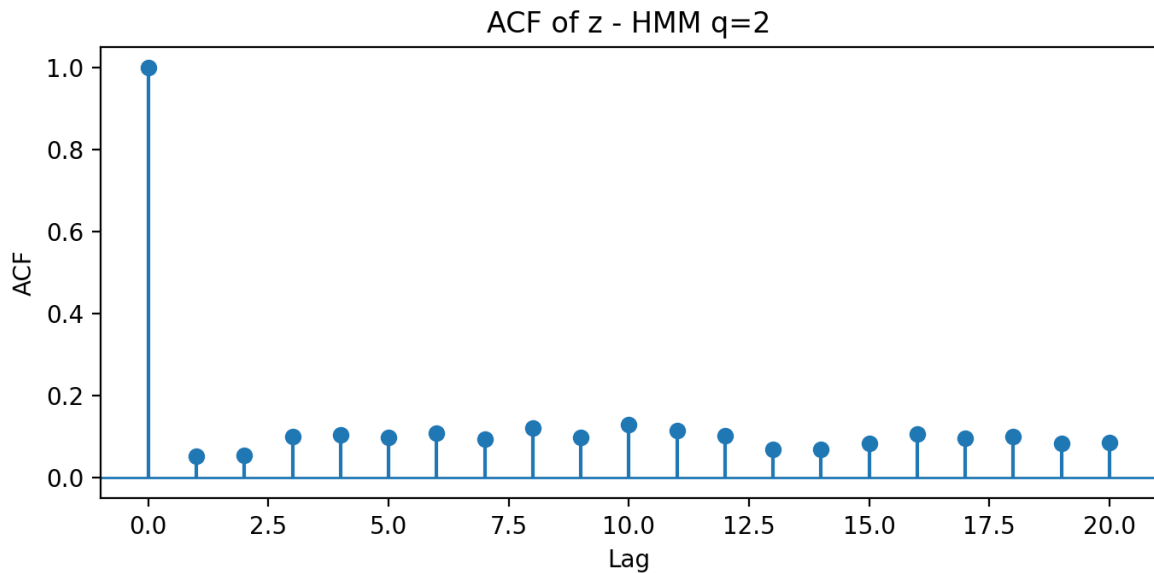


Figure 4. Q-Q plot of z - HMM $q=4$

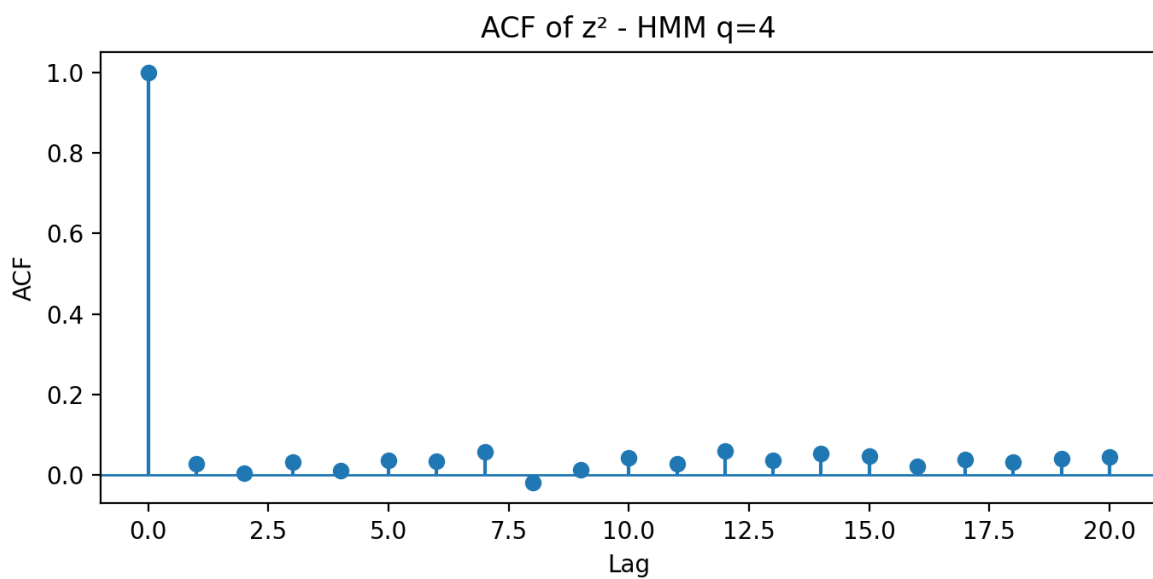
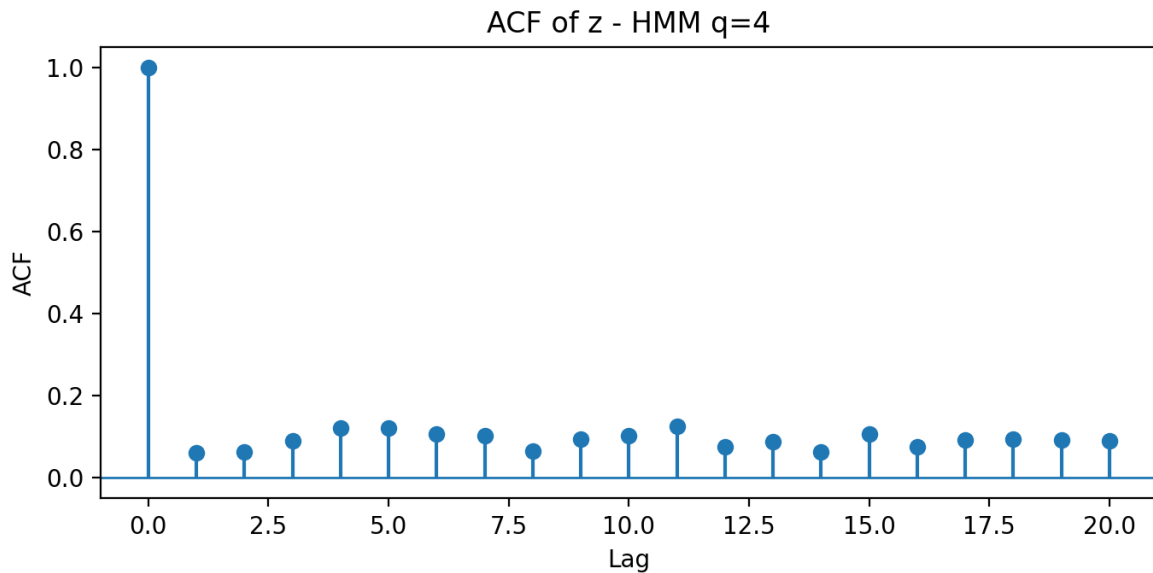
Autocorrelation diagnostics - HMM q=2

ACF plots complement Ljung-Box tests by visualizing serial dependence in z and z^2 .



Autocorrelation diagnostics - HMM q=4

Remaining dependence in z^2 indicates incomplete capture of volatility clustering.



Emission distribution and numerical stability

Table 2. Jarque-Bera normality rejections within regimes

Model	State	# tests	# reject ($p < 0.05$)	Min p-value	Max p-value
q=2	0	2	2	1.12e-14	3.57e-10
q=2	1	2	2	8.27e-06	0.0104
q=4	0	6	5	9.03e-24	0.0533
q=4	1	6	5	3.32e-15	0.0641
q=4	2	6	3	1.06e-24	0.9017

Interpretation: extensive rejection of normality indicates heavy tails and/or skew remain within regimes, limiting the adequacy of Gaussian emissions.

Figure 5. Jarque-Bera p-values by state/series (q=2)

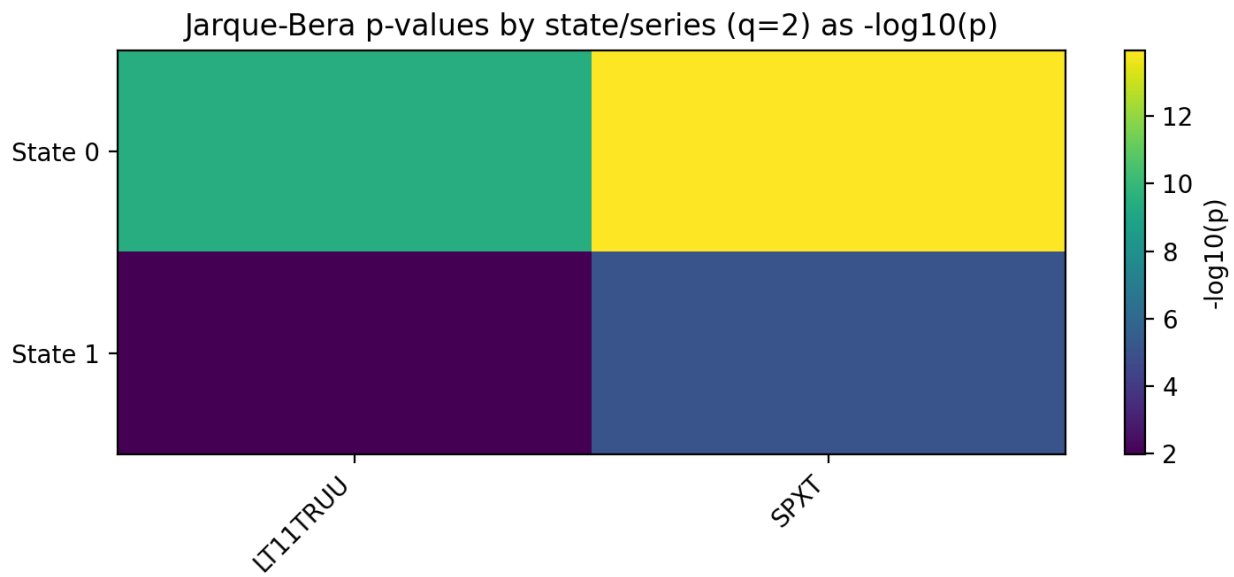


Figure 6. Jarque-Bera p-values by state/series (q=4)

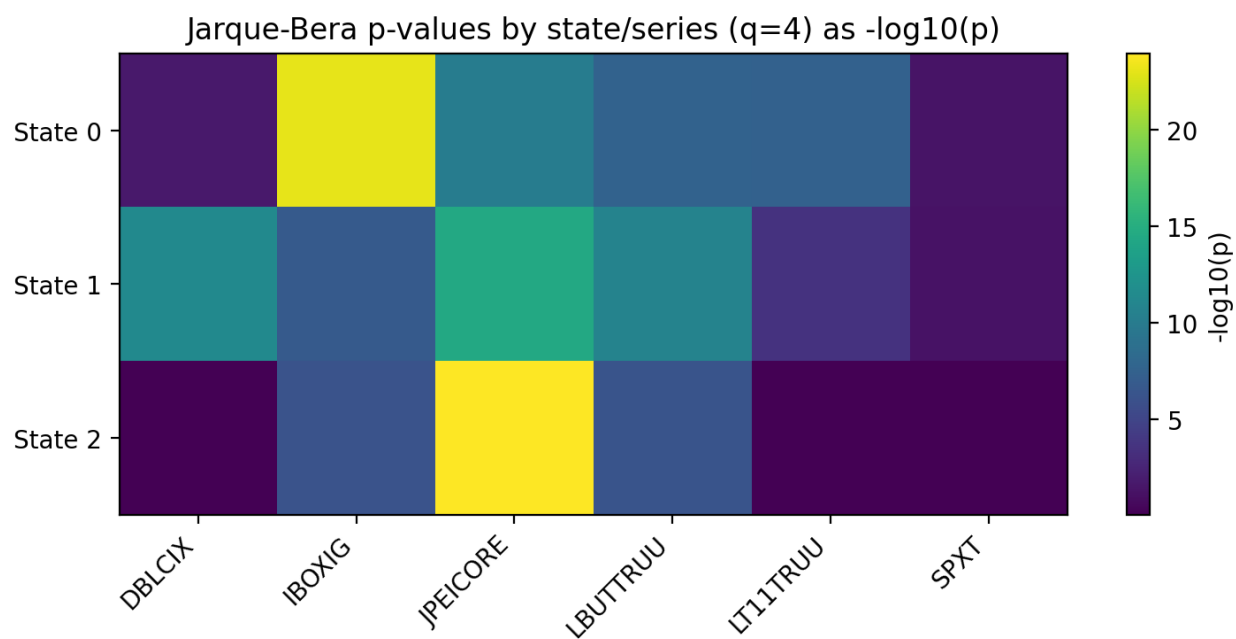
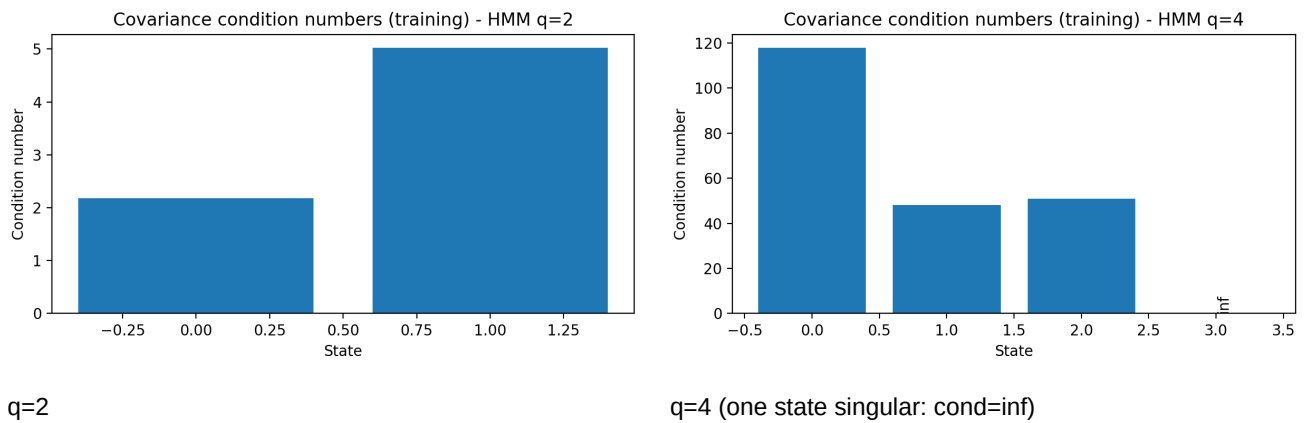


Figure 7. Covariance condition numbers (numerical stability)



A singular covariance in q=4 suggests weak identification or insufficient effective sample size in at least one regime; this undermines stable density evaluation.

Model selection: parametric bootstrap likelihood-ratio tests

Table 3. Bootstrap LR test results

Test	LR_obs	Bootstrap p-value	Bootstraps used	Failures	LL_null	LL_alt
K=1 Gaussian vs K=2 HMM (2D)	1134.57	0.01	198/200	2	14681.32	15248.61
K=1 Gaussian vs K=4 HMM (4D)	5197.05	0.01	195/200	5	51784.05	54382.57
K=2 HMM vs K=4 HMM (4D)	1136.55	0	200/200	0	53814.29	54382.57

Interpretation: observed LR statistics lie far in the upper tail of the bootstrap distributions, providing strong evidence in favor of additional regimes.

Figure 8. Bootstrap LR distribution - K=1 Gaussian vs K=2 HMM (2D)

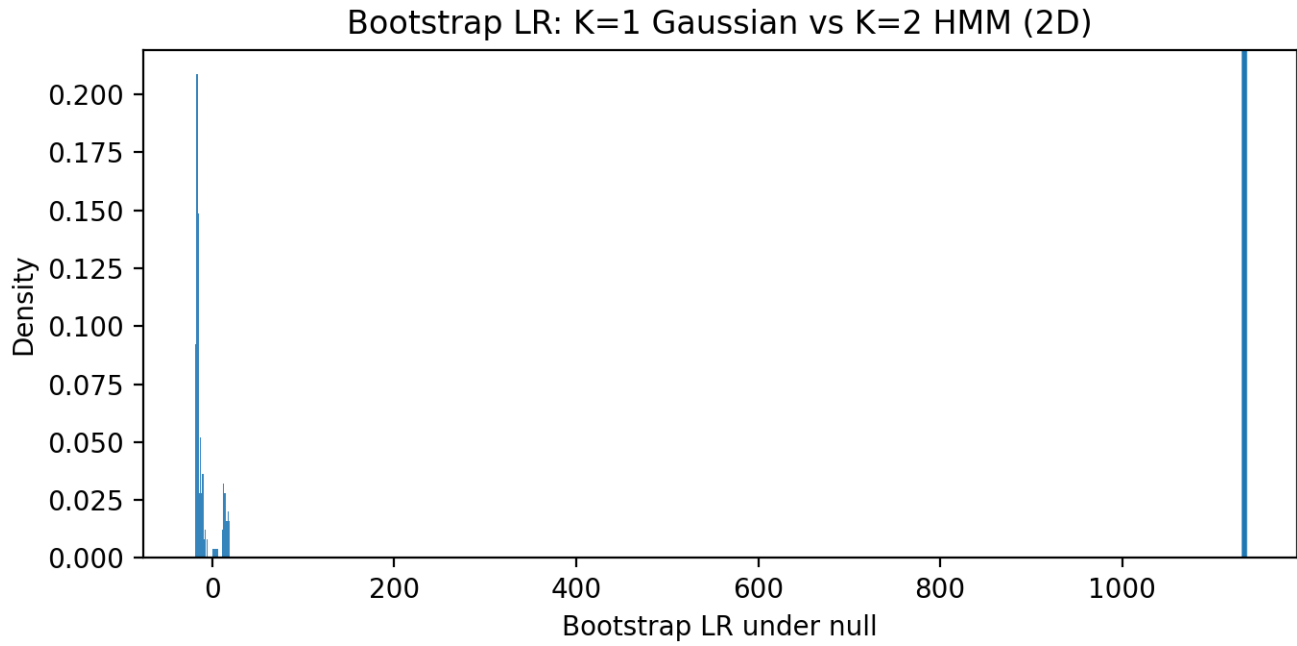


Figure 9. Bootstrap LR distribution - K=1 Gaussian vs K=4 HMM (4D)

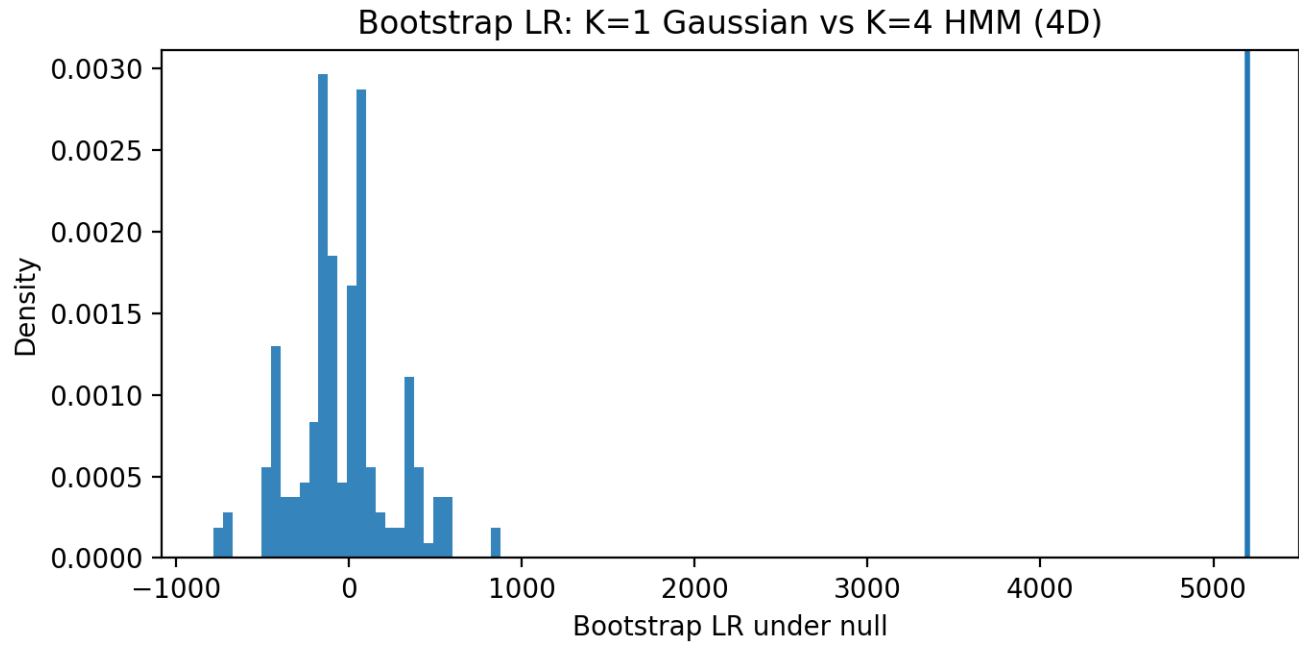
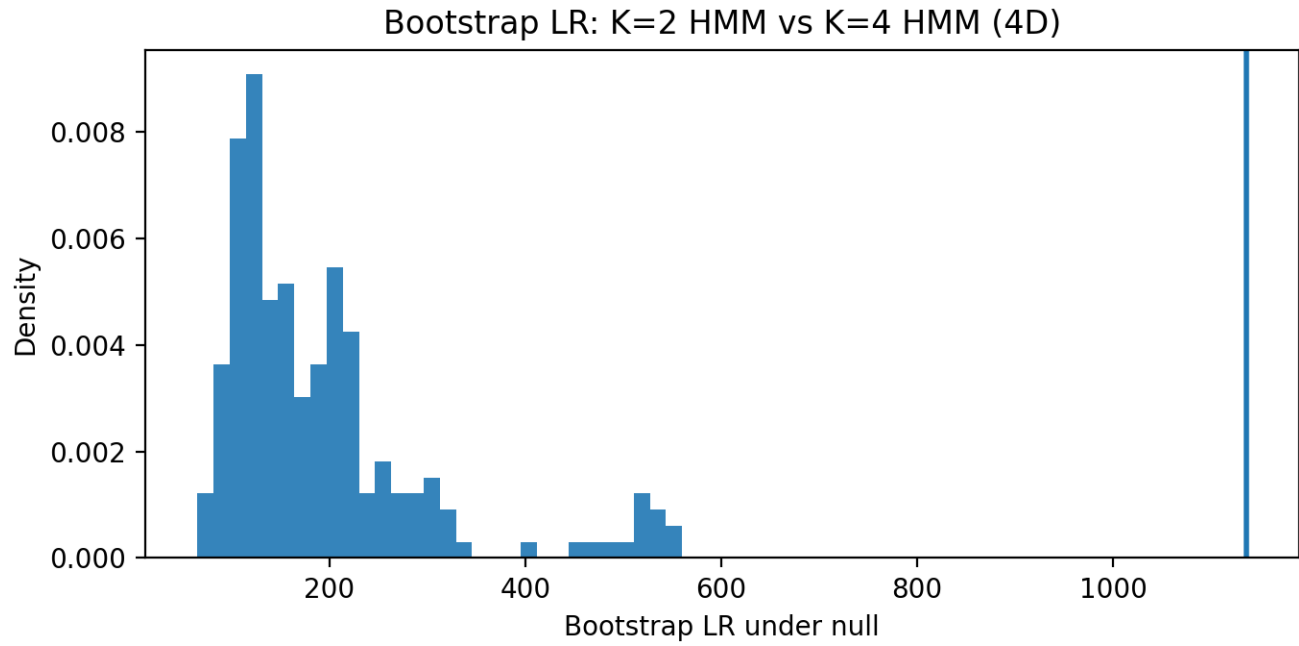


Figure 10. Bootstrap LR distribution - K=2 HMM vs K=4 HMM (4D)



Assessment of Phase 1 methodology soundness

Regime structure is supported. Bootstrap LR tests strongly favor richer regime structure:

- (i) K=1 Gaussian vs K=2 HMM (2D) yields LR_obs=1134.6 with bootstrap $p \approx 0.010$;
- (ii) K=1 Gaussian vs K=4 HMM (4D) yields LR_obs=5197.1 with bootstrap $p \approx 0.010$;
- (iii) K=2 HMM vs K=4 HMM (4D) yields LR_obs=1136.6 with p effectively below the bootstrap resolution.

However, the Gaussian HMM is not a well-calibrated density model. PIT uniformity is rejected for both $q=2$ and $q=4$, and transformed PITs exhibit strong autocorrelation (Ljung-Box p -values near zero). In $q=4$, remaining volatility clustering is also detected by ARCH LM ($p=0.0176$). Jarque-Bera tests reject normality within many regimes/series, particularly for credit-related series, indicating heavy tails and/or skew remain within states.

Numerical instability exists for $q=4$. One regime exhibits a singular covariance matrix (infinite condition number), which signals weak identification, inadequate effective sample size within that state, or near-collinearity within regime-conditional data.

Recommendations and path toward an HSMM with covariate-driven duration exit

If Phase 1 is used primarily for state labeling:

- The current approach can still be serviceable as a labeling mechanism, especially given strong evidence for regime structure.
- However, interpret state labels as heuristic segmentation rather than probabilistically calibrated regimes.

If Phase 1 must support probabilistic inference or defensible stylized-facts matching:

- 1) **Replace Gaussian emissions with heavier-tailed distributions** (e.g., Student-t) or robust mixtures. This directly targets Jarque-Bera rejections and PIT miscalibration due to heavy tails.
- 2) **Model time-varying volatility within regimes** (e.g., regime-switching GARCH or stochastic volatility). This targets the Ljung-Box on z^2 and ARCH LM rejections (remaining volatility clustering).
- 3) **Strengthen regime identifiability controls** for $q=4$ (e.g., increase the minimum regime size constraint, covariance flooring, or shrinkage) to prevent singular covariance states and unstable estimation.

Motivation for HSMM with covariate-dependent duration exit:

A standard HMM implies geometric state durations (memoryless exit probability). Financial regimes often exhibit non-geometric durations (e.g., crisis regimes persist and exit is driven by changing conditions). The strong residual dependence and evidence of state estimation instability are consistent with the possibility that regime persistence and exit are not well represented by a stationary transition matrix alone. An HSMM explicitly models the duration distribution D in each state, breaking the memoryless assumption. With covariates x_t , one can model the hazard of exit $h_i(t)$ or duration parameters as functions of x_t , allowing observable signals to influence the probability of leaving a regime.

Practical HSMM specification suggestion:

- Duration model: state i has duration distribution D_i with parameters $\theta_i(x_t)$, e.g., log-link: $\log E[D_i|x_t] = a_i + b_i' x_t$, or a discrete-time hazard: $\text{logit } P(\text{exit at time } t \mid \text{state } i, \text{ duration so far, } x_t) = a_i + c_i * f(\text{duration}) + b_i' x_t$.
- Candidate covariates: realized volatility, volatility indices (e.g., VIX proxy), drawdown measures, credit spreads, macro surprises, liquidity proxies, or cross-asset momentum/mean-reversion indicators.
- Validation: after implementing HSMM, re-run the same PIT and dependence diagnostics. A successful HSMM should improve PIT uniformity, reduce Ljung-Box rejections, and produce more stable regime-conditional covariance estimates.

Next diagnostic to confirm HSMM need directly:

Compute empirical state-duration distributions from the saved state path and compare them to the geometric durations implied by the estimated p_{ii} . Large discrepancies (especially excess long runs) provide direct statistical justification for HSMM.